

User Manual

Version 1.0 — October 2026

TDRP diagnosis, posterior-tree calibration, structural validation, and advanced utilities

Software: DeepPoW is an R/Shiny platform for diagnosing and calibrating the time-dependent rate phenomenon (TDRP) in viral evolution.

Primary purpose: DeepPoW complements, rather than replaces, upstream molecular-clock inference in BEAST and preliminary exploration in TreeTime.

License: MIT License

Project: https://github.com/BioEasy/DeepPoW

Important: DeepPoW operates on trees and posterior ensembles generated by upstream analyses. It does not repair poor sequence quality, recombination, or failed Bayesian convergence. Mode A and Mode B are complementary but methodologically distinct.

Content

1. INTRODUCTION AND SCOPE **1**

1.1 What DeepPoW does 1
1.2 What DeepPoW does not do 1

2. DEEPPOW WORKFLOW AT A GLANCE **1**

2.1 Mode A and Mode B are intentionally different 1

3. INSTALLATION AND ENVIRONMENT SETUP **2**

3.1 System requirements 2
3.2 Install R dependencies 2
3.3 Launch DeepPoW 2
3.4 Configure Foldseek and TM-align 2

4. INPUT DATA PREPARATION AND QUALITY CONTROL **3**

4.1 BEAST summary tree for Mode A 3
4.2 BEAST posterior ensemble for Mode B 3
4.3 TreeTime input 3
4.4 Structural data 3
4.5 Recombination and partitioning 3

5. MODE A — FAST DIAGNOSTICS **3**

5.1 Phenomenological model 4
5.2 Step-by-step operation 4
5.3 Robustness diagnostic 4
5.4 Traffic Light decision system 4
5.5 Why $\kappa > 150$ is screened in inter-host analyses 5
5.6 Expert Mode 5
5.7 AIC/BIC model comparison 5

6. MODE B — POSTERIOR-TREE CALIBRATION **5**

6.1 What Mode B uses 5
6.2 Step-by-step operation 6
6.3 Computational strategy 6
6.4 Output and posterior summarization 6

7. STRUCTURAL VALIDATION **6**

7.1 Input requirements 6
7.2 Step-by-step operation 6
7.3 Interpretation 6
7.4 Manuscript example 7

8. ADVANCED UTILITIES **7**

8.1 TreeTime TDRP Explorer 7
8.2 RMSD Matrix Generator 7
8.3 BEAST 2 XML Patcher 7

9. RECOMMENDED END-TO-END WORKFLOW **8**

9.1 Case-study patterns from the manuscript 8

10. TROUBLESHOOTING **8**

11. FREQUENTLY ASKED QUESTIONS **9**
12. INTERPRETATION AND REPORTING GUIDELINES **10**
12.1 EXAMPLE REPORTING SENTENCE 10
13. LIMITATIONS AND APPROPRIATE USE **10**
14. CITATION AND SUPPORT **10**

1. Introduction and scope

The time-dependent rate phenomenon (TDRP) describes the common observation that evolutionary rates estimated over short timescales can exceed long-term substitution rates. One important mechanistic explanation is incomplete purifying selection: slightly deleterious mutations may persist transiently in recently sampled populations and be removed over longer timescales. If short-term rates are extrapolated directly to deep divergence times, evolutionary histories can become artificially time-compressed.

DeepPoW converts the Prisoner of War (PoW) framework into an interactive R/Shiny workflow. The software combines two complementary engines: **Mode A** provides rapid diagnostic screening on a BEAST-derived summary tree, while **Mode B** performs mechanistic calibration across the full Bayesian posterior tree ensemble.

1.1 What DeepPoW does

- Fits a phenomenological rate-decay function to a time-scaled summary tree.
- Estimates the initial rate parameter (μ_0), long-term substitution rate (μ_∞), and decay coefficient (k).
- Performs robustness assessment by trimming the upper and lower 5% of extreme branch-rate observations and re-estimating parameters.
- Uses an automated Traffic Light system to distinguish robust, cautionary, and methodological-veto scenarios.
- Transforms Bayesian posterior trees using a mechanistic rate-class saturation framework.
- Uses monotonic numerical inversion and multi-core processing to accelerate posterior-tree transformation.
- Provides structural RMSD/Mantel-test utilities and TreeTime, RMSD-matrix, and BEAST 2 XML utilities.

1.2 What DeepPoW does not do

- It does not replace sequence alignment, phylogenetic inference, or BEAST MCMC.
- It does not guarantee that an upstream posterior is statistically valid.
- It does not automatically detect or resolve recombination.
- It does not establish that a structural correlation is direct proof of the PoW model.
- It does not make the $k > 150$ threshold a universal biological constant.

2. DeepPoW workflow at a glance

Stage	Input	DeepPoW action	Typical output
1. Upstream inference	Alignment + dates	BEAST / TreeTime	Time-scaled trees
2. Mode A	BEAST MCC/summary tree	NLS + robustness diagnostic	μ_0, μ_∞, k + Traffic Light
3. Decision	Mode A diagnostics	Assess signal and artifacts	Proceed / caution / veto
4. Mode B	BEAST posterior .trees	Mechanistic ensemble transformation	Calibrated posterior .trees
5. TreeAnnotator	Calibrated ensemble	Posterior summarization	Calibrated MCC tree
6. Structural Validation	Calibrated tree + RMSD	Mantel permutation test	r and P-value

2.1 Mode A and Mode B are intentionally different

Mode A uses a **phenomenological** approximation to diagnose whether a TDRP-like pattern is stable on a single summary tree. Mode B independently applies the mechanistic PoW rate-class saturation framework to posterior trees. The fitted k value from Mode A is therefore a diagnostic parameter and is not propagated as an input to the Mode B transformation.

Key rule: Mode A = Fast Diagnostics (phenomenological); Mode B = Posterior-Tree Calibration (mechanistic).

3. Installation and environment setup

3.1 System requirements

Component	Requirement / recommendation
Operating system	Windows, macOS, or Linux
R	R > 4.2.0 (as specified in the accompanying manuscript; verify against the installed release)
RStudio	Recommended
Browser	Modern browser supported by R/Shiny
Structural tools	Foldseek and/or TM-align for structural workflows
CPU	Multi-core CPU recommended for Mode B
Memory	Depends on taxa and posterior-tree ensemble size

3.2 Install R dependencies

Run the following in R/RStudio:

```
cran_packages <- c(
  "shiny", "bslib", "plotly", "DT", "shinycssloaders",
  "xml2", "ape", "nleqslv", "ggplot2", "shinyFiles",
  "minpack.lm", "tidytree", "stringr", "dplyr", "tidyr", "vegan"
)
new_packages <- cran_packages[
  !(cran_packages %in% installed.packages()[,"Package"])
]
if(length(new_packages)) install.packages(new_packages)

if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
if (!("treeio" %in% installed.packages()[,"Package"]))
  BiocManager::install("treeio")
```

3.3 Launch DeepPoW

1) Download and extract the DeepPoW repository.
2) Set the R working directory to the DeepPoW folder.
3) Launch the application with the command below.
4) Alternatively, use RStudio's green Run App button when DeepPoW.R is open.

```
setwd("path/to/DeepPoW")
shiny::runApp("DeepPoW.R")
```

3.4 Configure Foldseek and TM-align

Structural Validation and the RMSD Matrix Generator depend on external structural-alignment engines. The selected executable must be visible to the same operating-system environment used to launch R.

Linux/macOS with Conda:

```
conda install -c conda-forge -c bioconda foldseek
conda install -c bioconda tmalign
```

Verify installation before launching or restarting R/RStudio:

```
foldseek version
which foldseek
which TMalign
```

macOS Intel note: Use a Conda environment compatible with the installed Intel/x86_64 environment. If RStudio was opened before activating Conda, restart RStudio after changing PATH.

Windows releases may use the bundled executables in the DeepPoW plugin directory when those binaries are distributed with the release. Record the DeepPoW release and bundled tool versions for reproducibility.

4. Input data preparation and quality control

4.1 BEAST summary tree for Mode A

- Generate a maximum clade credibility (MCC) or otherwise appropriate posterior summary tree with TreeAnnotator.
- Use time-scaled node heights; branch lengths must represent evolutionary time rather than raw substitution distance.
- Retain the branch-specific evolutionary-rate information required by DeepPoW. If the required branch-rate annotations are missing, use the BEAST 2 XML Patcher (Section 8.3) when applicable.
- Export in a supported .nexus or .tree format.
- Keep tip labels unique and consistent with downstream structural data.

4.2 BEAST posterior ensemble for Mode B

- Provide the posterior tree ensemble (.trees) generated by BEAST.
- Remove the intended burn-in before DeepPoW processing (for example, 10%, according to the upstream analysis).
- Confirm adequate MCMC mixing and convergence before calibration.
- As a practical quality-control target, inspect Tracer; the manuscript case studies used ESS > 200 for relevant parameters.
- Use substitution-model assumptions consistent with the upstream BEAST analysis.

Important: DeepPoW cannot correct an unconverged or poorly mixed BEAST chain; upstream statistical deficiencies can be propagated into calibrated trees.

4.3 TreeTime input

Use a time-scaled ML tree, typically .nwk or .nexus. **TreeTime Explorer** is for rapid preliminary screening, not a substitute for Bayesian posterior calibration.

4.4 Structural data

- Prepare an All-vs-All structural RMSD matrix in CSV format, or generate one with the RMSD Matrix Generator.
- Matrix row names and column names must correspond exactly to phylogenetic tip labels.
- The matrix should contain pairwise structural distances with a zero diagonal.
- Protein structures can be supplied as .pdb or .cif files.
- Structure filenames should match the taxon identifiers used in the phylogenetic analysis.

4.5 Recombination and partitioning

The current framework does not automatically detect recombination. For viruses with substantial recombination, perform recombination-aware preprocessing or partitioning before TDRP analysis. Mosaic evolutionary histories can obscure interpretation of a single whole-genome rate-decay trajectory.

5. Mode A — Fast Diagnostics

Mode A is the diagnostic gateway. It fits a phenomenological rate-decay trajectory to a time-scaled summary tree using nonlinear least-squares (NLS), followed by a robustness assessment.

5.1 Phenomenological model

Mode A fits the following phenomenological expectation for the time-averaged substitution rate over an evolutionary timescale t. The decay coefficient is denoted k in the phenomenological model; it is distinct from the parameters of the mechanistic PoW formulation:

$$\bar{\mu}(t) = \mu_\infty + (\mu_0 - \mu_\infty)\frac{1 - e^{-kt}}{kt}$$

Here, μ_0 is the initial rate parameter (the limiting time-averaged rate as t approaches zero), μ_∞ is the long-term substitution rate, and k is the decay coefficient. When t is measured in years, k is expressed in year^{-1}. The characteristic timescale $1/k$ belongs to the underlying exponential decay kernel; it is not the e-folding time of the time-averaged rate trajectory. At $t = 0$, the expression is evaluated using its limit, $\bar{\mu}(0) = \mu_0$.

Parameter	Meaning	Interpretation
μ_0	Initial rate parameter	Limiting time-averaged rate as t approaches zero; a fitted parameter, not a direct observation
μ_∞	Long-term substitution rate	Asymptotic rate approached at long times
k	Decay coefficient	Controls the timescale of the underlying exponential decay kernel
$1/k$	Characteristic timescale of exponential kernel	$1/k$; not the e-folding time of the time-averaged rate curve

Do not impose k within [0, 1]: k depends on the time unit. For example, $k = 23.61$ per year corresponds to a characteristic exponential-kernel timescale of about 15.4 days.

5.2 Step-by-step operation

1) Open Tab 1: PoW Dynamics & Calibration.
2) Select Mode A (Fast Diagnostics).
3) Upload a time-scaled BEAST summary tree.
4) Run the phenomenological NLS fit.
5) Review μ_0, μ_∞, k, fitted trajectory, and model-comparison information.
6) Review the robustness diagnostic, which trims the upper and lower 5% of extreme branch-rate observations and refits the model.
7) Use the Traffic Light result to assess suitability for routine calibration. A Red Light is a methodological calibration veto, not a software-enforced lockout; exploratory Mode B analyses remain available at the user's discretion.

5.3 Robustness diagnostic

The robustness analysis removes the most extreme 5% at each tail (10% total) and refits the model. The k shift is calculated as: When the full-fit k is zero or numerically negligible, relative percentage shifts require cautious interpretation.

$$\text{Shift (\%)} = \frac{k_{\text{trimmed}} - k_{\text{full}}}{k_{\text{full}}} \times 100$$

Small shifts suggest limited sensitivity to the trimmed observations, whereas large shifts indicate that the fitted decay coefficient is sensitive to extreme branch-rate observations. The trimming diagnostic alone cannot identify the biological or sampling-related cause of instability.

5.4 Traffic Light decision system

Status	Primary trigger	Interpretation	Recommended action
Green Light	Shift < 20% and $k \leq 150$ in inter-host analyses, provided no other cautionary condition applies	Robust TDRP-like signal	Proceed to Mode B
Yellow Light	$\mu_0 \leq \mu\infty$, or 20% ≤ k shift < 100%, unless a Red Light condition applies	Flat/inverse or moderately unstable signal	Review sampling/biology; Mode B may be used cautiously when not vetoed
Red Light	k shift ≥ 100%, or k > 150 in inter-host analyses	Potentially unstable or biologically implausible fit	Methodological calibration veto; investigate potential artifacts before routine calibration. Exploratory Mode B analyses remain available.

Decision logic: Green: k shift < 20%; Yellow: 20% ≤ k shift < 100%; Red: k shift ≥ 100%. Other diagnostic conditions, including a flat/inverse fitted trajectory and the heuristic inter-host $k > 150$ screen, may alter the final classification. The threshold is a screening heuristic, not a universal biological law.

5.5 Why $k > 150$ is screened in inter-host analyses

Dense epidemiological sampling may produce many short terminal branches, potentially contributing to unstable branch-rate estimates and unusually large fitted k values. However, a large k does not, by itself, establish a sampling artifact.

With time measured in years, $k = 150$ per year corresponds to 1/150 years (approximately 0.0067 years, or 2.4 days). DeepPoW uses $k > 150$ as a heuristic warning threshold for potentially unstable fits for standard inter-host analyses.

5.6 Expert Mode

Some intra-host evolutionary processes can occur over very short timescales. Expert Mode bypasses the heuristic $k > 150$ screen and should be used only when the biological system and sampling design justify a substantially shorter temporal resolution. It does not resolve poor data quality or other robustness concerns.

Best practice: Record whether Expert Mode was enabled and explain the biological rationale.

5.7 AIC/BIC model comparison

Mode A provides comparative model information, including AIC/BIC-based comparisons of constant-rate and phenomenological rate-decay fits. Information criteria should be interpreted together with biological plausibility and robustness; a favorable BIC does not override a severe robustness failure or the inter-host $k > 150$ screen.

6. Mode B — Posterior-Tree Calibration

Mode B performs the mechanistic time-scale transformation across the full Bayesian posterior tree ensemble. It is the calibration engine and is deliberately separated from the phenomenological diagnostic fit in Mode A.

6.1 What Mode B uses

- Posterior trees from BEAST.
- Posterior evolutionary information, including mean substitution rates as available in the MCMC output.
- Substitution-model parameters required by the selected model.
- HKY85 transition/transversion ratio and stationary nucleotide frequencies when HKY85 is selected.
- JC69 assumptions when JC69 is selected.

Important: The Mode A k estimate is not used as an input to the Mode B mechanistic transformation.

6.2 Step-by-step operation

1) Open Tab 1: PoW Dynamics & Calibration and switch to Mode B (Posterior-Tree Calibration).
2) Upload the short-term substitution-rate log and the burn-in-removed BEAST posterior .trees file.
3) Select HKY85 or JC69 according to the upstream evolutionary model.
4) For HKY85, upload the HKY parameter log or supply the transition/transversion ratio and stationary nucleotide frequencies as required by the interface.
5) Set μ_{max} and tree sampling.
6) Validate the inputs and run posterior-tree calibration.
7) Download the calibrated posterior tree ensemble.

6.3 Computational strategy

Mode B numerically inverts the mechanistic PoW relationship between expected substitution distance and evolutionary time. DeepPoW constructs a monotonic mapping over a numerical grid and uses interpolation for rapid inverse lookup. Posterior trees are transformed independently using multi-core parallelization.

6.4 Output and posterior summarization

The calibrated ensemble is exported in a multi-tree (.trees) representation. Use TreeAnnotator to summarize the calibrated ensemble and generate the final calibrated MCC tree with posterior uncertainty, including 95% HPD intervals where supported by the annotation workflow. The 95% HPD intervals summarize posterior uncertainty; they are not themselves the targets of calibration.

7. Structural Validation

Structural Validation compares evolutionary distance derived from PoW-calibrated trees with pairwise protein structural divergence using an All-vs-All RMSD matrix and a Mantel permutation test.

7.1 Input requirements

- A post-calibration phylogeny with the required annotation.
- A pairwise RMSD matrix in CSV format.
- Exact agreement between matrix row/column identifiers and tree tip labels.
- A biologically corresponding protein/structure dataset.

7.2 Step-by-step operation

1) Open the Structural Validation tab.
2) Upload the annotated post-calibration phylogeny.
3) Upload the structural RMSD matrix.
4) Check that taxon names match exactly.
5) Set the number of permutations and run the Mantel test.
6) Report the Mantel *r* and permutation *P*-value together with the number of permutations and other analysis settings.

7.3 Interpretation

A positive Mantel association indicates that greater phylogenetic distance tends to be associated with greater structural divergence in the tested dataset.

Important limitation: Structural correlation is not direct proof of the PoW model. Protein structural divergence can be influenced by multiple factors beyond sequence-based evolutionary distance. Fix the random seed and record the number of permutations when reproducible Mantel *P*-values are required.

7.4 Manuscript example

The manuscript reports an independent analysis of 27 sobemovirus RdRp proteins representing 26 viral species, with a weak but significant positive association between PoW-calibrated phylogenetic distance and structural RMSD (Mantel $r = 0.162$, $P = 0.004$). This was treated as complementary structural evidence rather than direct validation of the PoW model.

8. Advanced Utilities

8.1 TreeTime TDRP Explorer

Provides rapid heuristic screening of a time-scaled ML tree and is intended as a lightweight exploratory step before intensive BEAST MCMC analysis.

1) Open Advanced Utilities → TreeTime Explorer.
2) Upload a TreeTime-generated .nwk or .nexus tree.
3) Click Run TDRP Explorer.
4) Review the fitted trajectory, run the sensitivity check, and inspect the Traffic Light status.

Interpretation: TreeTime Explorer is a preliminary screening tool, not a substitute for Bayesian posterior calibration.

8.2 RMSD Matrix Generator

1) Open Advanced Utilities → RMSD Matrix Generator.
2) Upload multiple .pdb or .cif structures.
3) Ensure filenames correspond exactly to tree taxon identifiers.
4) Select Foldseek or TM-align.
5) Click Generate RMSD Matrix.
6) Export the CSV matrix for Structural Validation. Review the execution trace and, where available, export the structural heatmap (PDF).

Engine PATH: Leave blank to auto-detect; if unsuccessful, select the executable from the bundled plugin directory (when available). Verify that the executable is compatible with the operating system.

8.3 BEAST 2 XML Patcher

The XML Patcher reduces manual BEAST 2 XML editing by inserting supported PoW-related model components and required state-node configurations.

1) Prepare the dataset in BEAUti and configure site models, clock models, and priors.
2) Save the unpatched XML.
3) Open Advanced Utilities → BEAST 2 XML Patcher.
4) Upload XML — Load the original BEAST 2 XML file.
5) Patch XML — Apply the required XML modifications.
6) Download Patched XML — Export the modified BEAST 2 XML file.
7) Run Patched XML with Statefile — Execute BEAST 2 using the patched XML and its compatible statefile.
8) Use TreeAnnotator to summarize the resulting posterior-tree ensemble and generate an MCC tree with branch-specific substitution-rate annotations for subsequent DeepPoW Mode A diagnostics.

Important: Ensure that the statefile is compatible with the patched XML, including its model configuration and state-node definitions. Verify the BEAST 2 execution and resulting tree annotations before downstream analysis.

9. Recommended end-to-end workflow

Step	Recommended practice
A. Data QC	Check alignment quality, sampling dates, duplicate/near-identical sequences, and possible recombination.
B. Preliminary exploration	Optionally use TreeTime Explorer for rapid temporal/rate screening.
C. Bayesian inference	Run BEAST with an appropriate clock/substitution model and assess convergence.
D. Mode A	Evaluate μ_0, μ_∞, k, robustness, and Traffic Light status.
E. Decision	Green → proceed; Yellow → investigate and proceed cautiously if not vetoed; Red → resolve the issue before calibration.
F. Mode B	Transform the posterior ensemble mechanistically.
G. Summarize	Use TreeAnnotator to generate the calibrated MCC tree and posterior intervals.
H. Validate	If appropriate, compare calibrated phylogenetic distances with independent structural distances.
I. Report	Document software version, upstream settings, diagnostic result, Expert Mode status, and validation settings.

9.1 Case-study patterns from the manuscript

Dataset	Observed behavior	Lesson
Influenza A virus (M-gene dataset)	Stable TDRP; $k = 23.61$; ~15.4-day characteristic timescale of the underlying exponential decay kernel; small shifts	Robust Green Light example
HCV	Flat/inverse summary-tree signal; constant-rate model favored	Summary-tree paradox; posterior calibration can remain informative
PVY	Dense sampling; 482.59% k shift after trimming; veto	Severe parameter instability; methodological calibration veto
HIV	Moderate 48.63% k shift	Intermediate-bias cautionary scenario

10. Troubleshooting

Problem	Likely cause	Recommended action
Tree cannot be loaded	Unsupported/invalid format or malformed annotation	Re-export valid Nexus/Newick and verify annotations
No rate information detected	Branch-rate annotation missing	Re-export the BEAST summary tree with required rate information
Robustness fit is unstable	Weak temporal signal or influential extreme branches	Inspect branch rates, sampling density, and upstream clock model
k is extremely large	Potentially unstable fitting; short branches or sampling structure may contribute	Inspect sampling design; apply the inter-host $k > 150$ screen
k is negative	Fitted trajectory does not support conventional positive decay	Inspect μ0 vs μ∞, fit, temporal signal, and data quality
Mode A passes but Mode B fails	Posterior/model incompatibility or poor upstream MCMC	Check .trees integrity, model settings, ESS, and annotations
Foldseek not found	Executable absent from PATH	Check bundled plugin executable first; otherwise verify the system PATH and Conda environment
TM-align not found	Executable absent or naming mismatch	Check bundled plugin executable or system PATH; restart R/RStudio after environment changes
RMSD matrix rejected	Taxon mismatch or invalid matrix	Match row/column names to tree tips;

		check diagonal
Mantel test weak/non-significant	Weak association or noisy structures	Check structure quality, matrix construction, and biological correspondence
Patched XML fails	Unsupported XML/package configuration	Inspect XML and verify BEAST 2/package compatibility; run a short test first

11. Frequently Asked Questions

Q1. Must *k* be between 0 and 1?

No. *k* is a decay constant and depends on the time unit. Values >1 are mathematically valid.

Q2. Why can *k* be 20, 30, 100, or >400?

Large *k* corresponds to a short characteristic time (approximately 1/*k*). In inter-host outbreak data it may also reflect ultra-short branches and dense sampling.

Q3. Why is *k* not passed from Mode A to Mode B?

Mode A *k* is a phenomenological diagnostic parameter; Mode B uses the mechanistic rate-class saturation framework and posterior substitution-model information.

Q4. Why did I get a Yellow Light?

Common causes are $\mu_0 \leq \mu_\infty$ or a parameter shift of at least 20% but less than 100%. Review sampling and summary-tree limitations; proceed cautiously when no veto is triggered.

Q5. Why did I get a Red Light?

A parameter shift ≥100% or $k > 150$ in standard inter-host analysis flags potentially unstable fitting that warrants investigation and triggers a methodological calibration veto. This is an advisory classification, not a software-enforced lockout.

Q6. Can I override the *k* > 150 screen?

Only with strong biological justification. Expert Mode is intended for genuine short-timescale systems such as appropriate intra-host analyses.

Q7. Why does TreeTime give a different k from BEAST?

TreeTime and BEAST can produce different trees, rates, and uncertainty structures; their estimates are not expected to be identical.

Q8. Why does Mode A show little TDRP signal for a deep-time dataset?

MCC summarization can smooth posterior rate heterogeneity. A flat summary-tree signal can therefore motivate examination of the full posterior ensemble.

Q9. What if my BEAST ESS is low?

Improve the upstream MCMC analysis before calibration. DeepPoW does not repair poor mixing.

Q10. Does structural correlation prove PoW is correct?

No. It provides complementary structural evidence that calibrated evolutionary distance has a measurable relationship with structural divergence.

12. Interpretation and reporting guidelines

For reproducible publications, report at least:

- DeepPoW version/release date.
- R and relevant external-tool versions when structural utilities are used.
- Upstream BEAST version and relevant BEAST packages.
- Substitution and clock models.
- Number of posterior trees and burn-in treatment.
- MCMC convergence/ESS assessment.
- Mode A μ_0, μ_∞, k, and robustness shifts.
- Traffic Light result and Expert Mode status.
- Mode B model parameters and computational settings where relevant.
- TreeAnnotator settings used to summarize the calibrated posterior.
- Structural matrix generation method and Mantel-test settings.

12.1 Example reporting sentence

The BEAST MCC tree was evaluated using DeepPoW Mode A. The fitted phenomenological model yielded $\mu_0 = 1.40 \times 10^{-3}$, $\mu_\infty = 2.31 \times 10^{-4}$ substitutions/site/year, and $k = 23.61$ year^{-1}. After trimming the upper and lower 5% of extreme branch-rate observations, shifts in μ_0, μ_∞, and k were 1.00%, 0.51%, and 6.89%, respectively, resulting in a Green Light classification. The BEAST posterior ensemble was subsequently transformed using Mode B.

13. Limitations and appropriate use

- Mode B accuracy depends on the quality and convergence of the upstream Bayesian posterior.
- The $k > 150$ threshold is a heuristic designed for typical inter-host RNA-virus timescales and is not universal.
- Systems with substantially different evolutionary kinetics may require a custom threshold or biologically justified Expert Mode.
- Current parsing requires appropriately annotated time-scaled trees; some structured-coalescent summary trees may lack required branch-rate annotations.
- Recombination is not automatically detected.
- Structural validation is complementary evidence, not a direct test of the PoW mechanism.
- DeepPoW complements BEAST and TreeTime rather than replacing them.

14. Citation and support

If DeepPoW is used in a publication, cite the associated DeepPoW publication and report the software version.

Project repository: https://github.com/BioEasy/DeepPoW

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